

## Question

The second-period elements **X** and element **Y** react at  $20^{\circ}\text{C}$  to obtain the common Lewis acid molecule **A**. When **A** passes through the crystal of **X** at  $1850^{\circ}\text{C}$  and a pressure lower than 133 Pa, a reactive gas **B** is obtained, which is isoelectronic with CO. **B** and **A** co-condense at  $-196^{\circ}\text{C}$  in a 2:1 stoichiometric ratio to obtain compound **C**. **C** is unstable and disproportionates above  $-50^{\circ}\text{C}$  to form the yellow compound **D** and **E** in a ratio of 1:2. **E** exists in two conformations in the solid and gaseous states. **D** has a similar structure and bonding mode to the hydride of element **X**. Reacting **D** with CO can cause symmetrical cleavage to obtain the adduct **F**, and there is a  $C_3$  axis in **F**.

According to the above description, which of the following statements is incorrect (including calculations, error within 1%)?

- A. All other options are incorrect
- B. **Y** element is a pale yellow gas
- C. The relative molecular mass of **C** is approximately 127
- D. There exist 3 chemical environments of **X** in **D**
- E. The simplest formula of **E** is  $\text{XY}_2$
- F. The mass fraction of C in **F** is 6.5%

## Answer

Correct Answer: **A**

## Detailed Explanation

For a second-period element to form a common Lewis acid, then **A** can only be  $\text{BF}_3$ .

### CHECKPOINT

1 PTS

**A** is  $\text{BF}_3$

If **B** and  $\text{CO}$  are isoelectronic, then **B** can only be  $\text{BF}$ . Therefore, **X** is **B**, and **Y** is **F**. The simple substance of **Y** is a pale yellow gas, so **B** is correct.

### CHECKPOINT

1 PTS

**B** is  $\text{BF}$

### CHECKPOINT

1 PTS

**X** is **B**, **Y** is **F**

According to the question stem "**B** and **A** co-condense at  $-196^\circ\text{C}$  with a stoichiometric ratio of 2:1 to obtain compound **C**", **C** should be  $\text{B}_3\text{F}_5$ , with a relative molecular mass of approximately 127, so **C** is correct.

## CHECKPOINT

1 PTS

**C** is  $\text{B}_3\text{F}_5$

Since "**E** exists in two conformations in the solid and gaseous states", it can be determined that **E** is  $\text{B}_2\text{F}_4$ , and the two conformations are planar and trigonal pyramidal, respectively. The simplest formula is  $\text{XY}_2$ , so E is correct.

## CHECKPOINT

1 PTS

**E** is  $\text{B}_2\text{F}_4$

According to the question stem "**D** has a similar structure and bonding mode to the hydride of element **X**", there should be a three-center two-electron bond in **D**. Under this condition, **B** has 3 forms of existence: terminal  $\text{BF}_2$ , bridging  $\text{BF}_2$ , and central **B**. Therefore, for any structure  $\text{B}_a(\text{BF}_2)_b$ , **F** must be an even number. Let there be  $x$  terminal  $\text{BF}_2$  groups, which should satisfy  $4a = x + 2(b - x)$ , i.e.,  $4a = 2b - x$ . It should be ensured that  $0 < 2a < b$  and  $x < b$  and  $a < b - x$ , i.e.,  $0 < 2a < b < 3a$ .

To balance the equation with a product stoichiometric ratio of 1:2, when the reactant stoichiometric number is 2, **D** is  $\text{B}_2\text{F}_2$ , i.e.,  $\text{B}_0(\text{BF}_2)_2$ , which cannot be satisfied; when the reactant stoichiometric number is 4, **D** is  $\text{B}_8\text{F}_{12}$ , i.e.,  $\text{B}_2(\text{BF}_2)_6$ , which is feasible; when the reactant stoichiometric number is 6, **D** is  $\text{B}_{14}\text{F}_{22}$ , i.e.,  $\text{B}_3(\text{BF}_2)_{11}$ , which cannot be satisfied; when the reactant stoichiometric number is 8, **D** is  $\text{B}_{20}\text{F}_{32}$ , i.e.,  $\text{B}_4(\text{BF}_2)_{16}$ , which cannot be satisfied; higher stoichiometric numbers also cannot satisfy the conditions. Therefore, **D** is  $\text{B}_8\text{F}_{12}$ , i.e.,  $\text{B}_2(\text{BF}_2)_6$ , which has 2 central **B** atoms, 4 terminal  $\text{BF}_2$  groups, and 2 bridging  $\text{BF}_2$  groups, and there are 3 kinds of chemical environments of **X**, so D is correct.

CHECKPOINT

2 PTS

**D** is  $\text{B}_8\text{F}_{12}$ , i.e.,  $\text{B}_2(\text{BF}_2)_6$ , which has 2 central B atoms, 4 terminal  $\text{BF}_2$  groups, and 2 bridging  $\text{BF}_2$  groups, and there are 3 kinds of chemical environments of **X**.

According to the question stem "**F** has a  $C_3$  axis", it can be directly determined that **F** is  $\text{B}(\text{BF}_2)_3\text{CO}$ , and the mass fraction of C is 6.5%, so F is correct. Therefore, choose A.

CHECKPOINT

1 PTS

**F** is  $\text{B}(\text{BF}_2)_3\text{CO}$